



IT Operation Support System (ITOSS)

Complete User Manual

Version: 1.0

Document Date: February 2026

System: ITOSS – IT Operation Support System (ITOSS)

Prepared for:

Dashen Bank

Information Technology Department

Addis Ababa, Ethiopia

Document ID: ITOSS-UM-2026-001

Classification: Internal Use Only

Document Control

Version	Date of release	Author	Description	Approved By
1.0	2026-01-15	Seid Mohammed	Initial Draft	

Contact Information

User	Department	Email	Phone
System Administrator	IT Infrastructure	Seidm2031@gmail.com	+251985192541

Document Owner:

IT Infrastructure Department
Dashen Bank Head Office
Bole Road, Addis Ababa
P.O. Box 127552

Contact Address ; Whatsapp: +251 911575827 ; Fax: +251 11 557 8428 ; Addis Ababa, Ethiopia ; Swift: DASHETAA ; Email: info@dashenbanksc.com

This document is proprietary to Dashen Bank. Unauthorized reproduction, distribution, or disclosure of this material is strictly prohibited. All rights reserved.

Table of Contents

Table of Contents

.....	1
IT Operation Support System (ITOSS)	1
Complete User Manual.....	1
Document Control.....	1
Contact Information	1
Table of Contents	3
Section 1: Introduction	12
1.1 Purpose of This Manual	12
1.2 System Overview	12
1.3 ITOSS - Key System Features & Modules	12
1.4 Target Audience	13
1.5 How to Use This Manual	13
Section 2: Getting Started	14
2.1 System Requirements	14
Workstation Requirements	14
2.2 Browser Compatibility	14
2.3 Login Procedure.....	14
2.4 First-Time Login	14
2.5 Password Reset.....	15
2.6 Dashboard Overview	15
2.7 Navigation Menu.....	16
2.8 User Profile Management.....	17
2.9 Logout	17
2.10 System Availability	18
Section 3: Dashboard.....	19
3.1 Dashboard Layout	19
3.2 Statistics Cards.....	19
3.3 Recent Activity Feed	19
3.4 Quick Action Buttons	20
3.5 Charts and Graphs	20

3.6 ATM Status Overview	20
3.7 Computer Inventory Summary	20
3.8 Branch Distribution Map	21
3.9 Notification Center	21
3.10 Customizing Dashboard View	21
Section 4: ATM Management	22
4.1 ATM Overview	22
4.2 Viewing ATMs	22
4.3 ATM List View	22
4.4 ATM Details View	23
4.5 Adding a New ATM	24
4.6 Editing ATM Information	24
4.7 ATM Types and Models	25
4.8 ATM Locations	25
4.9 ATM Network Configuration	25
4.10 ATM Status Management	25
4.11 ATM Reports	26
4.12 Creating ATM Reports	26
4.13 Downtime Reasons	26
4.14 Action Taken Options	27
4.15 Closing ATM Reports	27
4.16 ATM Report History	27
4.17 ATM Search and Filters	28
4.18 ATM Bulk Operations	28
4.19 ATM Export Functions	28
Section 5: Computer Inventory	29
5.1 Computer Overview	29
5.2 Viewing Computers	29
5.3 Computer List View	29
5.4 Computer Details View	30
5.5 Adding a New Computer	30
5.6 Editing Computer Information	31
5.7 Computer Models	31

5.8 Hardware Types	32
5.9 Computer Specifications	32
5.10 Operating Systems	32
5.11 Antivirus Management	33
5.12 IP Address and Hostname Management	33
5.13 Computer Status	33
5.14 Warranty Management	33
5.15 Monitor Assignment	34
5.16 Computer Transfer Process	34
5.17 Computer Assignment to Users	34
5.18 Computer Maintenance Tracking	34
5.19 Computer Disposal.....	35
5.20 Computer History	35
5.21 Search and Filters	36
5.22 Bulk Operations	36
5.23 Export Functions.....	36
Section 6: Branch Management.....	37
6.1 Branch Overview	37
6.2 Branch Hierarchy	37
6.3 District Offices.....	37
6.4 Viewing Branches.....	38
6.5 Branch List View.....	38
6.6 Branch Tree View	38
6.7 Adding a New Branch.....	38
6.8 Editing Branch Information	39
6.9 Branch Grades	39
6.10 Branch Codes and Tags.....	40
6.11 Parent-Child Relationships	40
6.12 Closing Branches.....	40
6.13 Reopening Branches	40
6.14 Branch Search	41
6.15 Branch Reports	41
6.16 Branch Assets Overview	41

Section 7: Asset Management	42
7.1 Asset Management Overview.....	42
7.2 Asset Classes	42
7.3 Asset Assignments	42
7.4 Assigning Assets to Users	42
7.5 Returning Assets	43
7.6 Assignment History.....	43
7.7 Asset Transfers.....	44
7.8 Creating Transfer Requests	44
7.9 Approving Transfers	44
7.10 Transfer Types	45
7.11 Asset Maintenance	45
7.12 Creating Maintenance Records	45
7.13 Maintenance Status Tracking.....	46
7.14 Maintenance Approval Workflow	46
7.15 Asset Disposal	46
7.16 Disposal Workflow.....	46
7.17 Disposal Reasons	47
7.18 Asset Lifecycle Management.....	47
7.19 Asset Audit Trail.....	47
7.20 Asset Reports.....	47
Section 8: User Administration.....	48
8.1 User Management Overview	48
8.2 User Roles and Permissions	48
8.3 User Groups.....	48
8.4 Viewing Users.....	48
8.5 Adding New Users.....	49
8.6 Editing User Information	49
8.7 Activating/Deactivating Users.....	50
8.8 Password Management.....	50
8.9 Role Assignment.....	50
8.10 Permission Matrix.....	51
8.11 Creating Roles	51

8.12 Editing Role Permissions.....	51
8.13 Creating User Groups.....	51
8.14 Assigning Users to Groups	52
8.15 Group Permissions.....	52
8.16 User Activity Monitoring.....	52
8.17 User Search and Filters	52
8.18 User Export Functions.....	53
8.19 Login History.....	53
8.20 Session Management.....	53
Section 9: Activity Monitoring.....	54
9.1 Activity Log Overview	54
9.2 Viewing Activity Log	54
9.3 Activity Log Details.....	54
9.4 Event Types	54
9.5 Subject Types	55
9.6 Filtering Activity Log	55
9.7 Date Range Selection	55
9.8 User Filters	55
9.9 Subject Filters.....	56
9.10 Activity Log Export	56
9.11 Activity Log Archiving	56
9.12 Audit Trail.....	56
9.13 Compliance Reporting.....	56
Section 10: Reporting	58
10.1 Reporting Overview.....	58
10.2 Activity Reports	58
10.3 Creating Activity Reports	58
10.4 Task Management.....	59
10.5 Deliverables	59
10.6 Report Status.....	59
10.7 ATM Reports.....	59
10.8 ATM Downtime Analysis	60
10.9 Computer Inventory Reports.....	60

10.10 Branch Asset Reports	60
10.11 Asset Valuation Reports	61
10.12 Maintenance Reports	61
10.13 Transfer Reports	61
10.14 Disposal Reports	62
10.15 User Activity Reports	62
10.16 Custom Reports	62
10.17 Report Export Options	63
10.18 Report Scheduling	63
10.19 Report Sharing	63
10.20 Dashboard Widgets	64
Section 11: Task Management	65
11.1 Task Overview	65
11.2 Task Categories	65
11.3 Creating Tasks	65
11.4 Assigning Tasks	66
11.5 Task Status	66
11.6 Task Priority	66
11.7 Due Dates	66
11.8 Task Dependencies	67
11.9 Task Comments	67
11.10 Task Attachments	67
11.11 Task Completion	67
11.12 Task History	68
11.13 Task Reports	68
11.14 Task Dashboard	68
Section 12: System Administration	69
12.1 System Configuration	69
12.2 Settings Management	69
12.3 Lookup Tables	69
12.4 ATM Downtime Reasons	70
12.5 Action Taken Options	70
12.6 Computer Models	70

12.7 Hardware Types	71
12.8 Asset Classes	71
12.9 Problem Categories	71
12.10 Branch Grades	71
12.11 District Management	72
12.12 System Backup	72
12.13 System Restore	72
12.14 Maintenance Mode	73
12.15 System Logs	73
12.16 Error Logs	73
12.17 Performance Monitoring	74
12.18 Cache Management	74
12.19 Session Management	74
12.20 Audit Settings	74
Section 13: Integration	75
14.1 Integration Overview	75
14.2 Active Directory Integration	75
14.3 Email Integration	75
14.4 SMS Integration	76
14.5 Notification Services	76
14.6 API Access	76
14.7 Third-Party Integrations	76
14.8 Data Import	77
14.9 Data Export	77
Section 14: Troubleshooting	77
15.1 Common Issues	77
15.2 Login Problems	78
15.3 Performance Issues	78
15.4 Data Not Saving	78
15.5 Report Generation Errors	79
15.6 Export Failures	79
15.7 ATM Report Issues	79
15.8 Computer Transfer Errors	79

15.9 Permission Errors	80
15.10 Activity Log Not Loading	80
15.11 Mobile Access Issues	80
15.12 Browser Compatibility Issues	80
15.13 Network Connectivity	81
15.14 Database Errors	81
15.15 Error Messages.....	81
15.16 Error Codes Reference	81
15.17 Escalation Procedures.....	82
15.18 Support Ticket Creation.....	82
Section 15: Best Practices	83
16.1 Data Entry Standards	83
16.2 Naming Conventions.....	83
16.3 Tag Number Formats	83
16.4 Regular Updates	83
16.5 Asset Tracking.....	83
16.6 Report Generation Schedule.....	84
16.7 User Training	84
16.8 Security Best Practices.....	84
16.9 Password Policies	84
16.10 Session Management.....	84
16.11 Data Backup	85
16.12 Audit Compliance	85
16.13 Performance Optimization.....	85
16.14 Regular Maintenance.....	85
Section 16: Appendices	86
17.1 Appendix A: Field Formats.....	86
17.2 Appendix B: Status Definitions.....	86
17.3 Appendix C: ATM Type Codes	86
17.4 Appendix D: Computer Models.....	86
17.5 Appendix E: Branch Codes	87
17.6 Appendix F: District Reference	87
17.7 Appendix G: Permission Matrix.....	88

17.8 Appendix H: Error Codes	88
17.9 Appendix I: Keyboard Shortcuts.....	89
17.10 Appendix J: Glossary of Terms	89
17.11 Appendix K: Frequently Asked Questions	90
17.12 Appendix L: Release Notes.....	90
17.13 Appendix M: System Requirements.....	90
17.14 Appendix N: Training Materials	91
17.15 Appendix O: Quick Reference Cards.....	91
Document Information	92

Section 1: Introduction

1.1 Purpose of This Manual

This IT Operation Support System (ITOSS) User Manual is designed to provide comprehensive guidance for all users of the Dashen Bank EMS platform. The manual serves as both an introductory guide for new users and a reference document for experienced users. It covers all aspects of the system including ATM management, computer inventory, branch management, asset tracking, reporting, and user administration.

1.2 System Overview

IT Operation Support System (ITOSS) is a centralized asset and infrastructure management platform developed for Dashen Bank. It enables the organization to efficiently track, monitor, and manage IT assets across all branches and departments. The system provides real-time visibility into the status and location of ATMs, computers, and other IT equipment, facilitating better decision-making and resource allocation.

System Name: IT Operation Support System (ITOSS)

Database: ems

Platform: Web-based Application

Developer: Dashen Bank IT Department

Launch Date: January 2026

1.3 ITOSS - Key System Features & Modules

Module	Description
Dashboard	Real-time overview with Active ATM Incidents and Digital Inventory Classification by district
Assets Management	Complete physical asset tracking including Computers, Printers, Scanners, Network Switches, Dongles, Monitors, Photocopiers, and Non-digital assets
Asset Audit	Asset lifecycle management with Audit Logs, Custody History, Disposals, Transfers, and Computer Histories
Channels	Terminal and channel device management for ATMs, POS machines, TM Downtime, Dashen Cards, and DOBs
Working Units	Organizational hierarchy management for Branches, Districts, Departments, Outlets, and Office Structure
Reporting	Operational reporting including District Activity, Inventory, Monthly ATM, Daily ATM, Weekly/Monthly, and Activity Reports
Settings	Master data configuration for Asset Classes, Computer Models, Hardware Types, Vendors, Problem Categories, Downtime Reasons, Task Categories, and more
User Management	Role-based access control with Users, User Groups, Permissions, and Online User monitoring
PM (Preventive Maintenance)	Maintenance scheduling and management with Create PM Plan, Branch PM Item, and PM Item Setting

Support	Repair tracking and access requests through Computer Maintenance and Get Passes
Administration	Central administration via Asset Manager and Report Manager
Utilities	System utilities including ATM Report API Test, Export Database, and System Configuration
Subscriptions	Subscription services management for Data VPNs and Fixed Lines
Search	Global search functionality across all modules and records

Key Capabilities: Asset Lifecycle Management • Real-time Monitoring • Audit Trail • Role-Based Access • Comprehensive Reporting • Maintenance Planning

1.4 Target Audience

This manual is intended for:

- **IT Administrators** – System configuration and user management
- **Branch Managers** – Branch asset oversight
- **Technicians** – ATM and computer maintenance
- **Asset Managers** – Asset lifecycle management
- **Auditors** – Compliance and activity monitoring
- **End Users** – General system usage

1.5 How to Use This Manual

This manual is organized into logical sections. Users should:

1. **New Users:** Start with Section 2 (Getting Started)
2. **Daily Operations:** Refer to Sections 4-7 for operational tasks
3. **Administrators:** Focus on Sections 8-12 for administration
4. **Troubleshooting:** Use Section 15 for issue resolution
5. **Quick Reference:** Appendices provide quick reference materials

Section 2: Getting Started

2.1 System Requirements

Workstation Requirements

Component	Minimum	Recommended
Processor	Intel Core i3 / AMD equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	100 MB free space	500 MB free space
Display	1366 x 768	1920 x 1080
Network	1 Mbps	5 Mbps or higher

2.2 Browser Compatibility

EMS supports the following browsers:

Browser	Minimum Version	Status
Google Chrome	90+	Fully Supported
Mozilla Firefox	88+	Fully Supported
Microsoft Edge	90+	Fully Supported
Safari	14+	Supported
Opera	76+	Supported

2.3 Login Procedure

1. Open your web browser
2. Navigate to the EMS URL: <https://itoss.dashenbanksc.com>
3. Enter your **Username**
4. Enter your **Password**
5. Click the **Login** button

2.4 First-Time Login

When logging in for the first time:

1. Use credentials provided by your system administrator
2. You will be prompted to change your password
3. Create a strong password following the policy:
 - Minimum 8 characters

- At least one uppercase letter
 - At least one lowercase letter
 - At least one number
 - At least one special character
4. Set your security questions (optional)
 5. Review and accept the terms of use

2.5 Password Reset

If you forget your password:

1. Click **Forgot Password** on the login screen
2. Enter your registered email address
3. Check your email for a password reset link
4. Click the link (valid for 24 hours)
5. Enter your new password
6. Confirm your new password
7. Click **Reset Password**

Alternatively, contact your system administrator for assistance.

2.6 Dashboard Overview

After successful login, you will see the main dashboard:

Dashboard Components:

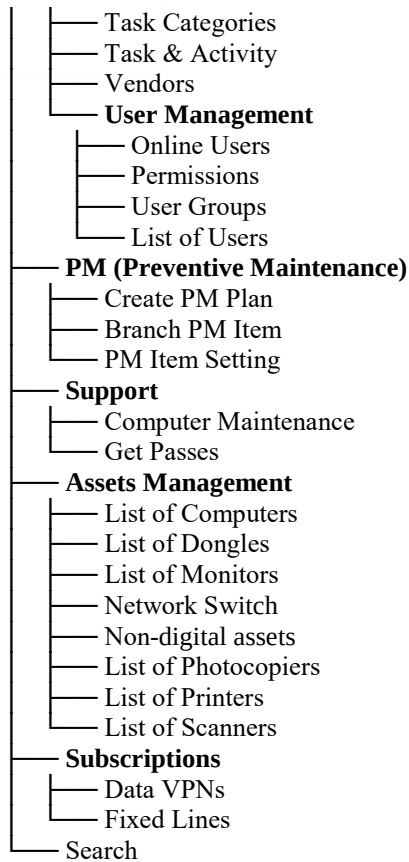
Component	Description
Header	User menu, notifications, search
Sidebar	Main navigation menu
Statistics Cards	Key performance indicators
Activity Feed	Recent system activity
Quick Actions	Common task shortcuts
Charts	Visual data representations
Footer	System information, version

2.7 Navigation Menu

The sidebar navigation menu provides access to all system modules:

ITOSS

- **Dashboard**
- **Home**
- **Administration**
 - Asset Manager
 - Report Manager
- **Utilities**
 - ATM Report API Test
 - Export Database
 - System Configuration
- **Asset Audit**
 - Audit Logs
 - Asset Custody History
 - Asset Disposals
 - Asset Transfers
 - Computer Histories
- **Reporting**
 - District Activity Report
 - Inventory Report
 - Monthly ATM Report
 - Monthly Report
 - Weekly Report
 - Daily ATM Reports
 - Activity Reports
- **Working Units**
 - Office Structure
 - List of Branches
 - List of Districts
 - Departments
 - List of Outlets
 - Organization
- **Channels**
 - TM Downtime
 - List of ATMs
 - Dashen Cards
 - List of DOBs
 - POS Machine
- **Settings**
 - Action Taken
 - Asset Classes
 - Card Categories
 - Computer Models
 - CPU Speeds
 - Deliverables
 - Downtime Reasons
 - Hardware Types
 - POS Models
 - Problem Categories
 - Problems
 - Quarters



2.8 User Profile Management

To manage your profile:

1. Click your name/avatar in the top-right corner
2. Select **Profile**
3. View and edit:
 - Personal information
 - Contact details
 - Password change
 - Notification preferences
 - Security settings
4. Click **Save Change**

2.9 Logout

To securely logout:

1. Click your name/avatar in the top-right corner

2. Select **Logout**
3. Confirm logout (if prompted)

The system will end your session and redirect to the login screen.

2.10 System Availability

EMS is available:

Schedule	Availability
Weekdays	24/7
Weekends	24/7
Holidays	24/7
Maintenance	Scheduled Sundays 2:00 AM - 4:00 AM

Emergency maintenance will be communicated via email notification.

Section 3: Dashboard

3.1 Dashboard Layout

The dashboard is your central hub for system information and quick access to common tasks.

Layout Structure:

- **Top Bar:** User menu, notifications, global search
- **Left Sidebar:** Main navigation
- **Main Content Area:** Statistics, charts, activity feed
- **Right Panel:** Quick actions, recent items (optional)

3.2 Statistics Cards

Statistics cards provide at-a-glance metrics:

Card	Description
Total ATMs	Count of all ATMs in system
Online ATMs	ATMs currently operational
Offline ATMs	ATMs currently down
Total Computers	Count of all computers
Active Branches	Operational branches
Open Reports	Pending activity reports
Assets Under Maintenance	Items in repair
Pending Transfers	Assets awaiting transfer

3.3 Recent Activity Feed

The activity feed displays the latest system actions:

Displayed Information:

- Timestamp
- User name
- Action type (created, updated, deleted)
- Subject type (ATM, Computer, Branch)
- Subject name/ID
- Link to view details

3.4 Quick Action Buttons

Common tasks available from the dashboard:

- **Add ATM** – Quick access to ATM creation
- **Add Computer** – Quick access to computer creation
- **Create Report** – New activity report
- **Transfer Asset** – Initiate asset transfer
- **View All Reports** – Reports list
- **Help** – Access documentation

3.5 Charts and Graphs

Visual representations of key data:

- **ATM Status Pie Chart** – Online vs offline distribution
- **Computer Inventory by Branch** – Bar chart
- **Asset Transfers Trend** – Line chart (monthly)
- **Reports by Status** – Donut chart
- **Branch Distribution** – Geographic map

3.6 ATM Status Overview

Quick view of ATM operational status:

- **Green:** Online/Operational
- **Red:** Offline/Downtime
- **Yellow:** Under Maintenance
- **Gray:** Disposed/Inactive

3.7 Computer Inventory Summary

Summary of computer assets:

- Total count
- By operating system
- By branch
- Warranty status
- Antivirus status

3.8 Branch Distribution Map

Geographic visualization of branches:

- **Markers** indicate branch locations
- **Color coding** by branch grade
- **Hover** for branch details
- **Click** to navigate to branch

3.9 Notification Center

Access notifications by clicking the bell icon:

Notification Types:

- System announcements
- Task assignments
- Report approvals
- Asset transfers
- Maintenance reminders
- System alerts

3.10 Customizing Dashboard View

To customize your dashboard:

1. Click the **Settings** (gear) icon on dashboard
2. Select widgets to display
3. Arrange widget order
4. Set refresh intervals
5. Save as default view

Section 4: ATM Management

4.1 ATM Overview

The ATM Management module provides comprehensive tracking of all Automated Teller Machines across Dashen Bank branches and off-branch locations.

Key Capabilities:

- Track ATM inventory with detailed specifications
- Monitor ATM operational status
- Record and manage downtime incidents
- Generate ATM performance reports
- Manage ATM transfers and disposals

4.2 Viewing ATMs

To view ATMs:

1. Navigate to **ATMs** in the sidebar
2. Select **All ATMs**
3. View the list of all ATMs

4.3 ATM List View

The ATM list displays:

Column	Description
ID	System identifier
Terminal	Terminal ID (e.g., PAGRL242)
Name	ATM name/location
OS	Operating system
Type	ATM type (GRG, NCR, GRG-Recycler)
Location	On-branch/Off-branch
Branch	Associated branch
IP Address	Network address
Status	Online/Offline
Actions	Edit, View, Report

4.4 ATM Details View

Click on any ATM to view detailed information:

General Information:

- Terminal ID
- ATM Name
- Operating System
- Type
- Location Type
- Design (Lobby/TTW)
- Branch

Network Configuration:

- IP Address
- Port
- Network Type (Broadband/SDWAN/4G)

Physical Attributes:

- Has Stabilizer (Yes/No)
- Is Relocating (Yes/No)
- Is Disposed (Yes/No)

Status Information:

- Is Online
- Last Checked At
- Remark

4.5 Adding a New ATM

To add a new ATM:

1. Navigate to **ATMs** → **Add ATM**
2. Complete the following fields:

Field	Required	Description	Example
Terminal	Yes	Unique terminal identifier	PAGRL242
Name	Yes	ATM name/location	Arbaminch Menariya ATM
OS	Yes	Operating system	Windows 10 Enterprise 64-bit
Type	Yes	ATM manufacturer/type	GRG
Location	Yes	Location category	Off-branch
Design	Yes	Physical design	Lobby
Branch	Yes	Associated branch	Arbaminch Branch (ID: 869)
IP Address	Yes	Network address	172.22.141.90
Port	Yes	Communication port	8003
Network Type	Yes	Network connection type	Broadband
Has Stabilizer	Yes	Stabilizer present	Checkbox
Is Relocating	No	Currently being moved	Checkbox
Remark	No	Additional notes	

3. Click **Save**
4. System confirms creation and logs in activity log

4.6 Editing ATM Information

To edit an ATM:

1. Navigate to **ATMs** → **All ATMs**
2. Find the ATM and click **Edit** (pencil icon)
3. Update necessary fields
4. Click **Update**
5. System logs changes in activity log showing before/after values

4.7 ATM Types and Models

ATM types in the system:

Type Code	Description	Examples
GRG	GRG Banking ATMs	PAGRL242, PAGRL191
NCR	NCR Corporation ATMs	PATML059, PATMT077
GRG-Recycler	GRG Recycler models	PAGRR823, PAGRR802

4.8 ATM Locations

Location categories:

Location	Description
On-branch	Located within branch premises
Off-branch	Located outside branch (merchant, public area)
Branch	Legacy location value (deprecated)
off-branch/Merchant	Merchant location

4.9 ATM Network Configuration

Network types:

Network Type	Description
Broadband	Standard broadband connection
SDWAN	Software-Defined WAN
4G	4G/LTE cellular
SD-WAN	Alternative SD-WAN designation

4.10 ATM Status Management

ATM status is tracked through:

- **is_online** – Real-time operational status (1=Online, 0=Offline)
- **last_checked_at** – Timestamp of last status check
- **isDisposed** – Permanently removed from service
- **isRelocating** – Currently being moved

4.11 ATM Reports

ATM reports track downtime incidents and resolution actions.

Report Status:

- **Open** – Incident reported, not yet resolved
- **Closed** – Incident resolved

4.12 Creating ATM Reports

To create an ATM downtime report:

1. Navigate to **ATMs** → **All ATMs**
2. Click **Create Report** for the affected ATM OR Navigate to **ATMs** → **ATM Reports** → **Create Report**
3. Complete the report form:

Field	Description
ATM	Select from dropdown
Downtime Reason	Select from list
Action Taken	Select from list
Open Date	Date/time of incident (auto-filled)
Call ID	Support call reference
TT	Trouble Ticket number
Remark	Additional notes

4. Click **Submit**
5. System creates report with status "Open"

4.13 Downtime Reasons

Available downtime reasons (from action_takens table):

ID	Name	Case Category
1	Power	Branch
2	Cash-out	Branch
3	Vendor Case	Vendor
4	Network case	Ethiotelcom
5	Relocation	Digital-Channel
6	Corridor dep't	Cyber-Security

4.14 Action Taken Options

Action taken options (from action_takens table):

ID	Name	Description
1	Under follow-up with the branch	Branch follow-up
2	Communicated with the Branch	Branch communication
3	Under follow-up with the vendor	Vendor follow-up
4	Under follow-up with Ethiotelcom	Telecom follow-up
5	Under follow-up with digital	Digital channel
6	Under follow-up with Cyber security	Security follow-up
7	Changed Host IP remotely and Restarted	Remote fix
8	Physically visit and Network cable replaced	On-site fix

4.15 Closing ATM Reports

To close an open report:

1. Navigate to **ATMs** → **ATM Reports**
2. Find the open report
3. Click **Close Report**
4. Confirm closure (optional: add resolution notes)
5. System updates:
 - **closed_by** – Current user ID
 - **close_date** – Current timestamp
 - Report status becomes closed

4.16 ATM Report History

View all reports for an ATM:

1. Navigate to **ATMs** → **All ATMs**
2. Click on an ATM to view details
3. Select the **Reports** tab
4. View all historical reports with:
 - Open/close dates
 - Downtime reason
 - Action taken
 - Created by/closed by

4.17 ATM Search and Filters

Search and filter ATMs using:

- Terminal ID
- Name
- Branch
- Type
- Location
- Status
- IP Address

4.18 ATM Bulk Operations

Perform actions on multiple ATMs:

1. Select checkboxes for multiple ATMs
2. Choose action:
 - Update Status
 - Transfer to Branch
 - Export Selected
 - Generate Report
3. Confirm action

4.19 ATM Export Functions

Export ATM data to:

- **Excel** (.xlsx)
- **CSV** (.csv)
- **PDF** (.pdf)

Section 5: Computer Inventory

5.1 Computer Overview

The Computer Inventory module manages all computer assets across the organization, including desktops, laptops, and servers.

Key Capabilities:

- Track computer hardware specifications
- Monitor operating systems and software
- Manage IP addresses and hostnames
- Track warranty and maintenance status
- Assign computers to users and branches

5.2 Viewing Computers

To view computers:

1. Navigate to **Computers** in the sidebar
2. Select **All Computers**
3. View the list of all computers

5.3 Computer List View

The computer list displays:

Column	Description
ID	System identifier
Tag No	Asset tag (e.g., DB/JM/4.1/17591)
Serial No	Manufacturer serial number
Model	Computer model
Hardware Type	Desktop/Laptop/Server
Branch	Assigned branch
Status	Operational/Non-Operational
IP Address	Network address
Host Name	Computer name
Actions	Edit, View, Transfer

5.4 Computer Details View

Click on any computer to view detailed information:

Hardware Information:

- Hardware Type
- Computer Model
- Tag Number
- Serial Number
- Hard Disk Size
- RAM Size
- Antivirus Status
- Operating System
- Activation Status

Network Information:

- IP Address
- Host Name

Assignment Information:

- Branch
- Has Monitor
- Under Maintenance
- Has Warranty

Status:

- Operational Status
- Deleted At (if soft-deleted)

5.5 Adding a New Computer

To add a new computer:

1. Navigate to **Computers** → **Add Computer**
2. Complete the following fields:

Field	Required	Description	Example
Hardware Type	Yes	Desktop/Laptop/Server	Desktop
Computer Model	Yes	Select from dropdown	HP ProDesk 600 G2
Tag No	Yes	Unique asset identifier	DB/JM/4.1/17591
Serial No	Yes	Manufacturer serial	YLT1A7C8

Hard Disk Size	Yes	Storage capacity	1TB
RAM Size	Yes	Memory size	8GB
Antivirus Status	Yes	Active Agent/Inactive	Active Agent
Operating System	Yes	OS version	Windows 11 Pro
Is Activated	Yes	Activation status	Checkbox
IP Address	Yes	Network address	192.168.163.99
Host Name	Yes	Computer name	W-BRN-JM-17591
Branch	Yes	Assigned branch	Jimma Branch (ID: 1)
Has Monitor	Yes	Monitor present	Checkbox
Under Maintenance	No	Maintenance status	Checkbox
Has Warranty	No	Warranty status	Checkbox
Status	Yes	Operational status	Operational

3. Click **Save**
4. System confirms creation and logs in activity log

5.6 Editing Computer Information

To edit a computer:

1. Navigate to **Computers** → **All Computers**
2. Find the computer and click **Edit**
3. Update necessary fields
4. Click **Update**
5. System logs changes in activity log showing before/after values

5.7 Computer Models

Computer models are stored in the computer_models table:

ID	Model Name	Description
11	HP ProDesk 400 G4	Desktop
13	HP ProDesk 600 G2	Desktop
60	HP EliteDesk 800 G4	Desktop
61	HP EliteBook 840 G6	Laptop
62	HP ProBook 450 G5	Laptop
63	HP Compaq Pro 6300	Desktop
65	HP Compaq 8200 Elite	Desktop
66	HP Compaq 6200 Pro	Desktop
67	HP Compaq 6000 Pro	Desktop
72	HP Pro 3400 Series	Desktop
88	HP Compaq dc5800	Desktop

5.8 Hardware Types

Hardware types are stored in the hardware_types table:

ID Name

- 1 Desktop
- 2 Laptop
- 3 Server
- 4 Workstation

5.9 Computer Specifications

Hard Disk Size common values:

- 250GB
- 500GB
- 1TB
- 2TB
- 256GB SSD
- 512GB SSD

RAM Size common values:

- 2GB
- 4GB
- 8GB
- 16GB
- 32G

5.10 Operating Systems

Common operating systems:

OS	Description
Windows 10 Pro	Professional edition
Windows 10 Enterprise	Enterprise edition
Windows 11 Pro	Professional edition
Windows 11 Enterprise	Enterprise edition
Windows 7 Embedded	Embedded systems
Windows Server 2012	Server OS

5.11 Antivirus Management

Antivirus status field indicates protection status:

- **Active Agent** – Antivirus installed and active
- **Active agent** – Legacy value (normalized to Active Agent)
- **Inactive** – Not active or not installed

5.12 IP Address and Hostname Management

IP Address Format:

- Internal: 172.16.x.x – 172.31.x.x
- Local: 192.168.x.x

Hostname Convention:

W-BRN-[BRANCH_CODE]-[TAG_NUMBER]

Examples:

- W-BRN-JM-17591
- W-BRN-TBM-10001
- W-BRN-ADO-1DWG

5.13 Computer Status

Status options:

Status	Description
Operational	Working normally
Non-Operational	Not working, needs attention
Under Maintenance	Being repaired
Disposed	Removed from inventory
In Transit	Being transferred

5.14 Warranty Management

Warranty tracking:

- **has_warranty** – 1 = Under warranty, 0 = Out of warranty
- Warranty expiration date (not currently tracked in schema)
- Recommended: Add warranty expiry field

5.15 Monitor Assignment

Monitors are tracked separately in the monitors table but linked to computers:

- **has_monitor** – 1 = Monitor assigned, 0 = No monitor
- Monitor details stored in monitors table

5.16 Computer Transfer Process

To transfer a computer:

1. Navigate to **Computers** → **All Computers**
2. Find the computer and click **Transfer**
3. Select **To Branch**
4. Select **Action**: transfer/handover/takeover
5. Enter **Remarks**
6. Click **Confirm Transfer**
7. System:
 - Updates branch_id
 - Creates transfer record in asset_transfers
 - Logs activity

5.17 Computer Assignment to Users

To assign a computer to a user:

1. Navigate to **Computers** → **All Computers**
2. Find the computer and click **Assign**
3. Select **User** from dropdown
4. Enter **Condition Out** (New/Used/Good)
5. Set **Assigned At** date
6. Click **Assign**
7. System:
 - Creates assignment in asset_assignments
 - Logs activity

5.18 Computer Maintenance Tracking

To record maintenance:

1. Navigate to **Computers** → **All Computers**
2. Find the computer and click **Maintenance**
3. Complete maintenance form:

Field	Description
Branch	Current branch
OU	Organizational Unit
Problem	Description of issue
Sent Date	Date sent for repair
Return Date	Expected/actual return
Status	SENT/RECEIVED/IN_PROGRESS/CLOSED
Approval Status	PENDING/ACCEPTED/REJECTED

4. Click **Save**

5.19 Computer Disposal

To dispose of a computer:

1. Navigate to **Computers** → **All Computers**
2. Find the computer and click **Dispose**
3. Enter **Reason** for disposal
4. Confirm
5. System:
 - Sets appropriate disposal flags
 - Records disposal in asset_disposals
 - Logs activity

5.20 Computer History

View complete history of a computer:

1. Open computer details
2. Click **History** tab
3. View chronological list of:
 - Creation
 - Updates (with before/after values)
 - Transfers
 - Assignments
 - Maintenance
 - Disposal

5.21 Search and Filters

Search computers using:

- Tag Number
- Serial Number
- Model
- Branch
- Status
- IP Address
- Host Name
- Operating System

5.22 Bulk Operations

Select multiple computers to:

- Update Status
- Transfer to Branch
- Assign to User
- Export Selected
- Generate Report

5.23 Export Functions

Export computer data to:

- **Excel** (.xlsx) – Full inventory
- **CSV** (.csv) – For data analysis
- **PDF** (.pdf) – For reporting

Section 6: Branch Management

6.1 Branch Overview

The Branch Management module maintains the organizational structure of all Dashen Bank branches and district offices.

Key Capabilities:

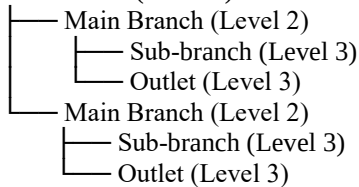
- Hierarchical branch structure
- District office management
- Branch grade classification
- Branch status tracking
- Parent-child relationships

6.2 Branch Hierarchy

Branches are organized hierarchically:

ITOSS

District Office (Level 1)



6.3 District Offices

District offices are regional administrative hubs:

ID	Name	Code
1	Jimma District Office	JDO
2	South West District Office	SWDO
3	Nekemte District Office	NDO
4	Hawasa District Office	HDO
5	Adama District Office	ADO
6	Dire Dawa District Office	DDDO
7	Mekele District Office	MDO
8	Dessie District Office	DDO
9	Bahir Dar District Office	BDO
10	North Addis District Office	NADO
11	East Addis District Office	EADO

12	West Addis District Office	WADO
13	South Addis District Office	SADO
14	Wolaita District Office	WDO
15	Head Office	HO

6.4 Viewing Branches

To view branches:

1. Navigate to **Branches** in the sidebar
2. Select **All Branches** or **Branch Tree**

6.5 Branch List View

The branch list displays:

Column	Description
ID	System identifier
Code	Branch code
Name	Branch name
Grade	Branch grade (I-V)
Tag	Branch tag (e.g., JM, AGR)
District	Parent district
Status	Open/Closed
Parent	Parent branch
Actions	Edit, View, Close

6.6 Branch Tree View

The tree view displays the hierarchical structure:

text

- ▶ Jimma District Office (ID: 32)
 - ▶ Jimma Branch (ID: 1)
 - ▶ Agaro Branch (ID: 2)
 - ▶ Limmugenet Branch (ID: 3)
 - ▶ Yebu Branch (ID: 4)

Click ▶ to expand/collapse branches.

6.7 Adding a New Branch

To add a new branch:

1. Navigate to **Branches** → **Add Branch**
2. Complete the following fields:

Field	Required	Description	Example
Code	Yes	Branch code	21
Name	Yes	Branch name	Jimma Branch
Grade	Yes	Branch grade	II
Tag	Yes	Branch tag	JM
District	Yes	Parent district	Jimma District (ID: 1)
Parent Branch	No	Parent branch (for sub-branches)	
isClosed	No	Closed status	Checkbox (default: 0)

3. Click **Save**
4. System confirms creation

6.8 Editing Branch Information

To edit a branch:

1. Navigate to **Branches** → **All Branches**
2. Find the branch and click **Edit**
3. Update necessary fields
4. Click **Update**
5. System logs changes in activity log

6.9 Branch Grades

Branch grades classification:

Grade	Description
I	Grade I Branch
II	Grade II Branch
III	Grade III Branch
IV	Grade IV Branch
V	Grade V Branch

Additional tier designations:

- Grade I Tier I
- Grade I Tier II
- Grade I Tier III

6.10 Branch Codes and Tags

Branch codes and tags are unique identifiers:

Branch	Code	Tag
Jimma Branch	21	JM
Agaro Branch	22	AGR
Limmugenet Branch	23	LMG
Arbaminch Branch	052	AM
Wolaita Sodo Branch	043	WST

6.11 Parent-Child Relationships

Sub-branches and outlets have parent relationships:

- **parent_id** references the main branch
- Used for hierarchical reporting
- Example: Outlet with parent_id = 1 (Jimma Branch)

6.12 Closing Branches

To close a branch:

1. Navigate to **Branches** → **All Branches**
2. Find the branch and click **Close**
3. Confirm closure
4. System sets **isClosed** = 1
5. Logs activity

Example:

text

```
UPDATE branches SET isClosed = 1 WHERE id = 1017;
```

6.13 Reopening Branches

To reopen a closed branch:

1. Navigate to **Branches** → **All Branches**
2. Filter by **Closed** status
3. Find the branch and click **Reopen**
4. Confirm
5. System sets **isClosed** = 0

6.14 Branch Search

Search branches using:

- Branch Code
- Branch Name
- Tag
- District
- Grade
- Status

6.15 Branch Reports

Generate branch reports:

1. Navigate to **Branches** → **Reports**
2. Select report type:
 - Branch Directory
 - Branch Assets Summary
 - Branch ATM Inventory
 - Branch Computer Inventory
3. Set filters
4. Click **Generate**
5. Export to Excel/PDF

6.16 Branch Assets Overview

View all assets at a branch:

1. Navigate to **Branches** → **All Branches**
2. Click on a branch
3. Select **Assets** tab
4. View:
 - ATMs at branch
 - Computers at branch
 - Other assets

Section 7: Asset Management

7.1 Asset Management Overview

The Asset Management module provides comprehensive lifecycle management for all organizational assets.

Key Capabilities:

- Asset classification
- Asset assignment to users
- Asset transfers between branches
- Maintenance tracking
- Disposal management
- Audit trail

7.2 Asset Classes

Assets are categorized into classes:

ID	Name	Description
1	Furniture & Office Materials	Office furniture and supplies
3	Electronics	Electronic equipment

7.3 Asset Assignments

Asset assignments track which user has which asset.

Assignment Status:

- **Active** – Currently assigned, no return date
- **Returned** – Asset returned, has return date

7.4 Assigning Assets to Users

To assign an asset:

1. Navigate to **Assets** → **Assignments**
2. Click **New Assignment**
3. Complete the form:

Field	Description
Asset Type	Computer/Monitor/Printer/etc.
Asset	Select specific asset
User	Select user to assign to
Assigned By	Automatically set to current user
Assigned At	Date/time of assignment
Condition Out	New/Used/Good/Fair

4. Click **Assign**
5. System creates assignment record

7.5 Returning Assets

To return an assigned asset:

1. Navigate to **Assets** → **Assignments**
2. Find active assignment
3. Click **Return**
4. Complete:

Field	Description
Returned By	Automatically set to current user
Returned At	Date/time of return
Condition In	New/Used/Good/Fair

5. Click **Confirm Return**
6. System updates assignment with return information

7.6 Assignment History

View assignment history for an asset:

1. Navigate to the asset (e.g., Computer)
2. Click **Assignments** tab
3. View all assignments with:
 - User
 - Assigned date
 - Returned date
 - Condition out/in

7.7 Asset Transfers

Asset transfers track movement of assets between branches.

Transfer Actions:

- **handover** – Hand over to another branch
- **takeover** – Take over from another branch
- **transfer** – Standard transfer
- **disposal** – Move to disposal

7.8 Creating Transfer Requests

To create a transfer:

1. Navigate to **Assets** → **Transfers**
2. Click **New Transfer**
3. Complete:

Field	Description
Asset Type	Type of asset
Asset	Select specific asset
From Branch	Auto-filled from asset
To Branch	Select destination
Action	transfer/handover/takeover/disposal
Performed By	Auto-filled
Remarks	Optional notes

4. Click **Create Transfer**

7.9 Approving Transfers

For transfers requiring approval:

1. Navigate to **Assets** → **Transfers Pending**
2. Review transfer details
3. Click **Approve** or **Reject**
4. Add comments if needed
5. Confirm

7.10 Transfer Types

Type	Description
transfer	Standard branch-to-branch transfer
handover	Handing over to another branch
takeover	Taking over from another branch
disposal	Moving to disposal process

7.11 Asset Maintenance

Track assets sent for repair or maintenance.

Maintenance Status:

- **SENT** – Sent for maintenance
- **RECEIVED** – Received back
- **IN_PROGRESS** – Currently being worked on
- **CLOSED** – Maintenance completed

Approval Status:

- **PENDING** – Awaiting approval
- **ACCEPTED** – Approved
- **REJECTED** – Rejected

7.12 Creating Maintenance Records

To create a maintenance record:

1. Navigate to **Assets** → **Maintenance**
2. Click **New Maintenance**
3. Complete:

Field	Description
Asset Type	Type of asset
Asset	Select specific asset
Branch	Current branch
OU	Organizational Unit
Problem	Description of issue
Sent Date	Date sent for repair
Status	SENT/RECEIVED/IN_PROGRESS/CLOSED
Approval Status	PENDING/ACCEPTED/REJECTED

4. Click **Save**

7.13 Maintenance Status Tracking

Update maintenance status:

1. Navigate to **Assets** → **Maintenance**
2. Find maintenance record
3. Click **Update Status**
4. Select new status
5. Add notes
6. Click **Update**

7.14 Maintenance Approval Workflow

For maintenance requiring approval:

1. Manager reviews pending maintenance
2. Click **Approve** or **Reject**
3. Comments required for rejection
4. Status updates accordingly

7.15 Asset Disposal

Track assets permanently removed from inventory.

Disposal Status:

- **disposed** – Permanently removed

7.16 Disposal Workflow

To dispose of an asset:

1. Navigate to **Assets** → **Disposals**
2. Click **New Disposal**
3. Complete:

Field	Description
Asset Type	Type of asset
Asset	Select specific asset
Branch	Current branch
Disposed By	Auto-filled
Disposed At	Date of disposal
Reason	Reason for disposal
Status	disposed (default)

4. Click **Confirm Disposal**
5. System updates asset status

7.17 Disposal Reasons

Common disposal reasons:

- End of life
- Beyond repair
- Obsolete
- Lost
- Stolen
- Donated

7.18 Asset Lifecycle Management

Complete asset lifecycle:

Text

Acquisition → Assignment → Transfer → Maintenance → Disposal

↑ ↓ ↓ ↓ ↓
 Register Assign to Move to Track Remove
 User Branch Repairs from System

7.19 Asset Audit Trail

Every asset action is logged:

- Creation
- Assignment
- Return
- Transfer
- Maintenance
- Disposal

7.20 Asset Reports

Generate asset reports:

- Asset Inventory by Branch
- Asset Assignment Report
- Transfer History
- Maintenance Summary
- Disposal Repor

Section 8: User Administration

8.1 User Management Overview

The User Administration module manages system access, roles, and permissions.

Key Capabilities:

- User account management
- Role-based access control
- Permission assignment
- User groups
- Activity monitoring

8.2 User Roles and Permissions

EMS uses role-based access control with granular permissions.

Permission Structure:

- Each permission follows: [subject].[action]
- Examples: atm.create, computer.update, branch.view

8.3 User Groups

Users can be organized into groups for easier permission management:

- Groups inherit permissions
- Users inherit group permissions
- Groups can be nested

8.4 Viewing Users

To view users:

1. Navigate to **Admin** → **Users**
2. View list of all user

User list displays:

Column	Description
ID	User identifier
Name	Full name
Email	Email address
Branch	Assigned branch
Status	Active/Inactive
Last Login	Last access time
Actions	Edit, Permissions, Disable

8.5 Adding New Users

To add a new user:

1. Navigate to **Admin** → **Users** → **Add User**
2. Complete the form:

Field	Required	Description
Name	Yes	Full name
Email	Yes	Email address (unique)
Username	Yes	Login username (unique)
Password	Yes	Initial password
Confirm Password	Yes	Password confirmation
Branch	Yes	Assigned branch
Is Active	Yes	Account status
Roles	No	Assign roles
User Groups	No	Assign to groups

3. Click **Save**
4. System creates user account
5. User receives welcome email (optional)

8.6 Editing User Information

To edit a user:

1. Navigate to **Admin** → **Users**
2. Find user and click **Edit**
3. Update fields
4. Click **Update**
5. Changes logged in activity log

8.7 Activating/Deactivating Users

To change user status:

1. Navigate to **Admin** → **Users**
2. Find user
3. Click **Activate** or **Deactivate**
4. Confirm
5. System updates **isActive** field

Example:

text

```
UPDATE users SET isActive = 0 WHERE id = 23;  
-- Logs: User 23 deactivated by user 1
```

8.8 Password Management

User self-service:

- Change password via Profile
- Forgot password via email

Administrator actions:

- Reset user password
- Send password reset link
- Force password change on next login

8.9 Role Assignment

To assign roles to a user:

1. Navigate to **Admin** → **Users**
2. Find user and click **Roles**
3. Select roles from list
4. Click **Assign**
5. User inherits role permissions

8.10 Permission Matrix

Permission structure (partial list):

Subject	Actions	Description
atm	create, update, delete, view, viewAny	ATM permissions
computer	create, update, delete, view, viewAny	Computer permissions
branch	create, update, delete, view, viewAny	Branch permissions
assetassignment	create, update, delete, view, viewAny	Asset assignment permissions
assettransfer	create, update, delete, view, viewAny	Transfer permissions
user	create, update, delete, view, viewAny	User management permissions

8.11 Creating Roles

To create a new role:

1. Navigate to **Admin** → **Roles**
2. Click **Add Role**
3. Enter:
 - Role Name
 - Description
4. Select permissions
5. Click **Save**

8.12 Editing Role Permissions

To edit role permissions:

1. Navigate to **Admin** → **Roles**
2. Find role and click **Permissions**
3. Check/uncheck permissions
4. Click **Save**
5. Changes apply to all users with this role

8.13 Creating User Groups

To create a user group:

1. Navigate to **Admin** → **User Groups**
2. Click **Add Group**
3. Enter:
 - Group Name
 - Description
4. Select permissions

5. Click **Save**

8.14 Assigning Users to Groups

To assign users to a group:

1. Navigate to **Admin** → **User Groups**
2. Find group and click **Members**
3. Select users from list
4. Click **Add to Group**
5. Users inherit group permissions

8.15 Group Permissions

Groups can have permissions assigned:

1. Navigate to **Admin** → **User Groups**
2. Find group and click **Permissions**
3. Select permissions
4. Click **Save**
5. All group members inherit these permissions

8.16 User Activity Monitoring

Monitor individual user activity:

1. Navigate to **Admin** → **Users**
2. Find user and click **Activity**
3. View chronological list of user actions
4. Filter by date range
5. Export activity report

8.17 User Search and Filters

Search users by:

- Name
- Email
- Username
- Branch
- Role
- Status
- Last login date

8.18 User Export Functions

Export user data to:

- **Excel** – User list
- **CSV** – For import/export
- **PDF** – User directory

8.19 Login History

View user login history:

1. Navigate to **Admin** → **Users**
2. Find user and click **Login History**
3. View:
 - Login timestamp
 - IP address
 - Browser/device
 - Success/Failure

8.20 Session Management

Manage active user sessions:

1. Navigate to **Admin** → **Sessions**
2. View all active sessions
3. Force logout specific users
4. Terminate all sessions (maintenance mode)

Section 9: Activity Monitoring

9.1 Activity Log Overview

The Activity Log provides a complete audit trail of all system actions.

Logged Information:

- Who performed the action
- What action was performed
- When it happened
- What changed (before/after)

9.2 Viewing Activity Log

To view activity log:

1. Navigate to **Admin** → **Activity Log**
2. View chronological list of activities

Activity log displays:

Column	Description
Timestamp	When action occurred
User	Who performed action
Log Name	Category (e.g., atm, computer)
Event	Action type (created/updated/deleted)
Subject	What was affected
Description	Summary of action
Actions	View details

9.3 Activity Log Details

Click on any log entry to view details:

9.4 Event Types

Event	Description
created	New record created
updated	Existing record updated
deleted	Record deleted

9.5 Subject Types

Subject	Description
ATM	ATM machine
Computer	Computer asset
Branch	Branch information
User	User account
ATM_report	ATM downtime report
AssetTransfer	Asset transfer record
AssetAssignment	Asset assignment

9.6 Filtering Activity Log

Filter options:

- Date range
- User
- Event type
- Subject type
- Log name

9.7 Date Range Selection

Select custom date ranges:

- Today
- Yesterday
- Last 7 days
- Last 30 days
- This month
- Last month
- Custom range

9.8 User Filters

Filter by specific user:

1. Click **User** filter
2. Select user from dropdown
3. View all actions by that user

9.9 Subject Filters

Filter by specific subject:

1. Click **Subject** filter
2. Select subject type
3. Optionally enter subject ID
4. View all actions for that subject

Example: Filter by subject_type = 'ATM' AND subject_id = 6

9.10 Activity Log Export

Export activity log to:

- **Excel** – For analysis
- **CSV** – For data processing
- **PDF** – For audit purposes

9.11 Activity Log Archiving

Automatic archiving:

- Logs older than 1 year automatically archived
- Archived logs available upon request
- Prevents performance degradation

9.12 Audit Trail

The activity log serves as the system's audit trail for compliance:

- SOX compliance
- Internal audit requirements
- Regulatory reporting
- Security investigations

9.13 Compliance Reporting

Generate compliance reports:

1. Navigate to **Reports** → **Compliance**
2. Select report type:
 - User Activity Summary
 - Data Change Report

- Access Log
 - Security Events
- 3. Set date range
- 4. Click **Generate**
- 5. Export for auditors

Section 10: Reporting

10.1 Reporting Overview

The Reporting module provides comprehensive reporting capabilities across all system modules.

Report Categories:

- Activity Reports
- ATM Reports
- Computer Inventory Reports
- Asset Reports
- Compliance Reports

10.2 Activity Reports

Activity reports track task completion and deliverables.

Fields:

- Task ID
- Deliverable ID
- Task Giver
- District
- User
- Status
- Description
- Report Date

10.3 Creating Activity Reports

To create an activity report:

1. Navigate to **Reports** → **Activity Reports**
2. Click **Create Report**
3. Complete:

Field	Description
Task	Select from dropdown
Deliverable	Select from dropdown
Task Giver	Person who assigned task
District	Associated district
User	Person completing task
Status	Completed/In Progress/Pending

Description	Detailed description
Report Date	Date of report

4. Click **Submit**

10.4 Task Management

Tasks are defined in the tasks table:

ID	Name	Description
1	PM Task	Preventive maintenance
13	Asset Audit	Annual asset audit
15	Network Review	Network infrastructure review
18	Security Check	Security compliance check

10.5 Deliverables

Deliverables are defined in the deliverables table:

ID	Name	Description
1	PM Report	Preventive maintenance report
13	Audit Report	Asset audit report
19	Security Report	Security assessment report

10.6 Report Status

Status options:

Status	Description
Completed	Task fully completed
In Progress	Task underway
Pending	Not yet started

10.7 ATM Reports

ATM reports track downtime incidents:

- Report ID
- ATM
- Downtime Reason
- Action Taken
- Open Date
- Close Date
- Call ID

- TT Number
- Created By
- Closed By

10.8 ATM Downtime Analysis

Analyze ATM downtime:

1. Navigate to **Reports** → **ATM Reports** → **Analysis**
2. Select date range
3. View:
 - Total downtime incidents
 - Average time to resolution
 - Top downtime reasons
 - Branch performance
 - ATM reliability metrics

10.9 Computer Inventory Reports

Generate computer inventory reports:

1. Navigate to **Reports** → **Computer Inventory**
2. Select filters:
 - Branch
 - Model
 - Status
 - OS Version
 - Warranty Status
3. Click **Generate**
4. View results
5. Export to Excel/PDF

Report includes:

- Total computers by branch
- Operating system distribution
- Warranty expiration summary
- Antivirus status
- Computers by model

10.10 Branch Asset Reports

Generate branch asset reports:

1. Navigate to **Reports** → **Branch Assets**

2. Select branch(es)
3. Select asset types
4. Click **Generate**
5. View:
 - ATMs at branch
 - Computers at branch
 - Other assets
 - Assignment status

10.11 Asset Valuation Reports

Generate asset valuation reports:

1. Navigate to **Reports** → **Asset Valuation**
2. Select asset classes
3. Set date range
4. Click **Generate**
5. View:
 - Original value
 - Depreciated value
 - Current value
 - By branch/district

10.12 Maintenance Reports

Generate maintenance reports:

1. Navigate to **Reports** → **Maintenance**
2. Select date range
3. Select asset types
4. Click **Generate**
5. View:
 - Maintenance history
 - Costs (if tracked)
 - Frequency by asset
 - Vendor performance

10.13 Transfer Reports

Generate transfer reports:

1. Navigate to **Reports** → **Transfers**
2. Select date range
3. Select from/to branches
4. Click **Generate**

5. View:
 - Transfer history
 - Transfer counts
 - Transfer types
 - Pending transfers

10.14 Disposal Reports

Generate disposal reports:

1. Navigate to **Reports** → **Disposals**
2. Select date range
3. Select disposal reasons
4. Click **Generate**
5. View:
 - Disposed assets
 - Disposal reasons
 - Disposal value

10.15 User Activity Reports

Generate user activity reports:

1. Navigate to **Reports** → **User Activity**
2. Select user(s)
3. Set date range
4. Click **Generate**
5. View:
 - Actions performed
 - Login history
 - Changes made

10.16 Custom Reports

Create custom reports using the report builder:

1. Navigate to **Reports** → **Custom Reports**
2. Click **Create Custom Report**
3. Select data source
4. Choose fields
5. Set filters

6. Define grouping
7. Choose chart type (optional)
8. Save report template
9. Generate on demand

10.17 Report Export Options

Export reports to:

Format	Use Case
Excel (.xlsx)	Data analysis
CSV (.csv)	Data import/export
PDF (.pdf)	Distribution/Printing
HTML (.html)	Web viewing
JSON (.json)	API consumption

10.18 Report Scheduling

Schedule reports for automatic generation:

1. Navigate to **Reports** → **Schedules**
2. Click **New Schedule**
3. Select report
4. Set frequency:
 - Daily
 - Weekly
 - Monthly
 - Quarterly
5. Set recipients
6. Choose format
7. Click **Save**

10.19 Report Sharing

Share reports with other users:

1. Generate report
2. Click **Share**
3. Select users/groups
4. Set permissions (view/edit)
5. Add message

6. Click **Share**

10.20 Dashboard Widgets

Add report widgets to dashboard:

1. Navigate to **Dashboard**
2. Click **Add Widget**
3. Select report type
4. Configure display
5. Click **Add**
6. Widget appears on dashboard

Section 11: Task Management

11.1 Task Overview

The Task Management module allows assignment and tracking of tasks and deliverables.

Key Capabilities:

- Task creation and assignment
- Due date tracking
- Status management
- Task dependencies
- Comments and attachments

11.2 Task Categories

Tasks are organized into categories:

Category	Description
Maintenance	Preventive/corrective maintenance
Audit	Internal/external audit tasks
Security	Security-related tasks
Deployment	New equipment deployment
Decommission	Asset decommissioning
Review	Periodic reviews
Report	Report generation

11.3 Creating Tasks

To create a task:

1. Navigate to **Tasks** → **All Tasks**
2. Click **Create Task**
3. Complete:

Field	Description
Title	Task title
Category	Task category
Description	Detailed description
Assigned To	User responsible
Task Giver	Person assigning
Priority	High/Medium/Low
Due Date	Expected completion

Deliverable	Expected output
Attachments	Supporting documents

4. Click **Save**

11.4 Assigning Tasks

Tasks can be assigned to:

- Individual users
- User groups
- Branches
- Districts

11.5 Task Status

Status	Description
Pending	Not yet started
In Progress	Work underway
Completed	Finished
Approved	Approved by manager
Rejected	Returned for revision
Cancelled	Cancelled

11.6 Task Priority

Priority	Description
High	Urgent, immediate attention
Medium	Normal priority
Low	Can be delayed

11.7 Due Dates

Due dates can be:

- Fixed date
- Relative to creation
- Recurring

Recurring options:

- Daily
- Weekly
- Monthly
- Quarterly
- Yearly

11.8 Task Dependencies

Tasks can have dependencies:

1. Navigate to task
2. Click **Dependencies**
3. Add dependent tasks
4. Set dependency type:
 - Blocked by
 - Blocks
 - Relates to

11.9 Task Comments

Add comments to tasks:

1. Navigate to task
2. Click **Comments**
3. Enter comment
4. Add attachments (optional)
5. Click **Post Comment**

11.10 Task Attachments

Upload attachments to tasks:

- Images
- Documents
- Spreadsheets
- PDFs

11.11 Task Completion

To mark a task as completed:

1. Navigate to task

2. Click **Complete**
3. Add completion notes
4. Upload deliverables
5. Click **Submit**
6. Notify task giver

11.12 Task History

View task history:

1. Navigate to task
2. Click **History**
3. View chronological list:
 - Status changes
 - Assignments
 - Comments
 - Attachments

11.13 Task Reports

Generate task reports:

- Tasks by user
- Tasks by branch
- Overdue tasks
- Completed tasks
- Task status summary

11.14 Task Dashboard

Task dashboard widgets:

- My Tasks
- Overdue Tasks
- Team Tasks
- Task Status Chart
- Upcoming Deadlines

Section 12: System Administration

12.1 System Configuration

System configuration options available to administrators:

1. Navigate to **Admin** → **Settings**
2. Configure:
 - System name
 - Logo
 - Timezone
 - Date format
 - Number format
 - Email settings
 - Session timeout
 - Password policy

12.2 Settings Management

Key settings categories:

Category	Settings
General	System name, logo, timezone
Security	Password policy, 2FA, session timeout
Email	SMTP settings, notifications
Notifications	Email, SMS, in-app
Reports	Export formats, max rows
Backup	Schedule, retention

12.3 Lookup Tables

Manage lookup tables:

1. Navigate to **Admin** → **Lookup Tables**
2. Select table to manage
3. Add, edit, or delete entries

12.4 ATM Downtime Reasons

Manage ATM downtime reasons:

ID	Reasons	Category
1	Power	Branch
2	Network	Ethiotelecom
3	Vendor Case	Vendor
4	Cash out	Branch
5	Note Jam	Branch
6	Relocation	Digital Channel
7	Corridor	Digital Channel
8	Rodent intrusion	Branch

12.5 Action Taken Options

Manage action taken options:

ID	Actions taken	Description
1	Under follow-up with the branch	Branch follow-up
2	Communicated with the Branch	Branch communication
3	Under follow-up with the vendor	Vendor follow-up
4	Under follow-up with Ethiotelecom	Telecom follow-up
5	Under follow-up with digital	Digital channel
6	Under follow-up with Cyber security	Security follow-up
7	Changed Host IP remotely and Restarted	Remote fix
8	Physically visit and Network cable replaced	On-site fix

12.6 Computer Models

Manage computer models:

ID	Name	Manufacturer
11	HP ProDesk 400 G4	HP
13	HP ProDesk 600 G2	HP
60	HP EliteDesk 800 G4	HP
61	HP EliteBook 840 G6	HP
62	HP ProBook 450 G5	HP
63	HP Compaq Pro 6300	HP
65	HP Compaq 8200 Elite	HP
66	HP Compaq 6200 Pro	HP
67	HP Compaq 6000 Pro	HP
72	HP Pro 3400 Series	HP
88	HP Compaq dc5800	HP

12.7 Hardware Types

Manage hardware types:

ID	Name
1	Desktop
2	Laptop
3	Server
4	Workstation

12.8 Asset Classes

Manage asset classes:

ID	Name	Description
1	Furniture & Office Materials	Office furniture and supplies
3	Electronics	Electronic equipment

12.9 Problem Categories

Manage problem categories for maintenance:

ID	Name	Description
1	Hardware Failure	Physical component failure
2	Software Issue	Operating system/application
3	Network Problem	Connectivity issues
4	Power Issue	Electrical/power problems
5	User Error	Operator mistake

12.10 Branch Grades

Manage branch grades:

Grade	Description
I	Grade I Branch
II	Grade II Branch
III	Grade III Branch
IV	Grade IV Branch
V	Grade V Branch

12.11 District Management

Manage districts:

ID	Name	Code
1	Jimma District Office	JDO
2	South West District Office	SWDO
3	Nekemte District Office	NDO
4	Hawasa District Office	HDO
5	Adama District Office	ADO
6	Dire Dawa District Office	DDDO
7	Mekele District Office	MDO
8	Dessie District Office	DDO
9	Bahir Dar District Office	BDO
10	North Addis District Office	NADO
11	East Addis District Office	EADO
12	West Addis District Office	WADO
13	South Addis District Office	SADO
14	Wolaita District Office	WDO
15	Head Office	HO

12.12 System Backup

Configure system backups:

1. Navigate to **Admin** → **Backup**
2. Configure:
 - Backup schedule
 - Backup location
 - Retention policy
 - Include/exclude tables
3. Manual backup option
4. Restore from backup

12.13 System Restore

Restore system from backup:

1. Navigate to **Admin** → **Restore**
2. Select backup file
3. Confirm restore
4. System enters maintenance mode
5. Restore database
6. Verify restoration

7. Exit maintenance mode

12.14 Maintenance Mode

Enable maintenance mode:

1. Navigate to **Admin** → **Maintenance**
2. Click **Enable Maintenance Mode**
3. Set message for users
4. Confirm
5. System unavailable to regular users
6. Administrators can still access

12.15 System Logs

View system logs:

1. Navigate to **Admin** → **System Logs**
2. Select log type:
 - Application log
 - Error log
 - Access log
 - Security log
3. View entries
4. Filter by date/level
5. Export logs

12.16 Error Logs

View application error logs:

- Timestamp
- Error level
- Error message
- File/line number
- Stack trace
- User (if available)

12.17 Performance Monitoring

Monitor system performance:

- Response time
- Database queries
- Memory usage
- CPU usage
- Active sessions
- Cache hit ratio

12.18 Cache Management

Manage system cache:

1. Navigate to **Admin** → **Cache**
2. View cache statistics
3. Clear specific cache:
 - Application cache
 - Route cache
 - Config cache
 - View cache
4. Clear all cache

12.19 Session Management

Manage user sessions:

1. Navigate to **Admin** → **Sessions**
2. View active sessions
3. Force logout specific users
4. Terminate all sessions
5. Configure session settings

12.20 Audit Settings

Configure audit settings:

- Log level
- Retention period
- Sensitive data masking
- Audit events to track

Section 13: Integration

14.1 Integration Overview

EMS integrates with various systems:

- Active Directory
- Email servers
- SMS gateways
- Monitoring systems
- Third-party APIs

14.2 Active Directory Integration

LDAP/Active Directory integration:

1. Navigate to **Admin** → **Integrations** → **LDAP**
2. Configure:
 - Server address
 - Port
 - Base DN
 - User attribute
 - Group mapping
3. Test connection
4. Enable synchronization

Benefits:

- Single sign-on
- Centralized user management
- Automatic group membership

14.3 Email Integration

Email integration for notifications:

1. Navigate to **Admin** → **Integrations** → **Email**
2. Configure SMTP:
 - Server
 - Port
 - Encryption
 - Authentication
 - From address
3. Test configuration

4. Enable email notifications

14.4 SMS Integration

SMS integration for alerts:

1. Navigate to **Admin** → **Integrations** → **SMS**
2. Configure provider:
 - Gateway URL
 - API key
 - Sender ID
3. Test message
4. Enable SMS notifications

14.5 Notification Services

Configure notification channels:

Channel	Use For
Email	Reports, assignments
SMS	Urgent alerts
In-app	All notifications
Push	Mobile alerts

14.6 API Access

EMS provides REST API for integration:

API Endpoints:

- GET /api/atms – List ATMs
- GET /api/atms/{id} – ATM details
- GET /api/computers – List computers
- GET /api/branches – List branches
- POST /api/reports – Create report

Authentication:

- API key
- OAuth2
- JWT tokens

14.7 Third-Party Integrations

Supported third-party integrations:

- **Monitoring tools** – Nagios, Zabbix
- **Help desk** – ServiceNow, Jira
- **Asset management** – SCCM, ManageEngine
- **Reporting** – Power BI, Tableau

14.8 Data Import

Import data from external sources:

1. Navigate to **Admin** → **Import**
2. Select import type:
 - ATMs
 - Computers
 - Branches
 - Users
3. Upload CSV/Excel file
4. Map columns
5. Validate data
6. Import
7. Review results

14.9 Data Export

Export data for external use:

1. Navigate to **Reports**
2. Generate report
3. Select export format
4. Download file
5. API access for automated exports

Section 14: Troubleshooting

15.1 Common Issues

Issue	Solution
Cannot login	Check credentials, reset password
Page not loading	Check network, clear cache
Data not saving	Check required fields, refresh
Report errors	Verify filters, reduce date range
Export fails	Reduce data size, try different format

15.2 Login Problems

Cannot login – incorrect password:

1. Click **Forgot Password**
2. Follow email instructions
3. Contact administrator

Account locked:

1. Wait 15 minutes for auto-unlock
2. Contact administrator

Account inactive:

1. Contact administrator to activate

15.3 Performance Issues

Slow page loading:

- Clear browser cache
- Reduce displayed records
- Use filters
- Check internet connection
- Contact IT if persistent

15.4 Data Not Saving

Form won't submit:

- Check required fields (*)

- Verify field formats
- Check for duplicate entries
- Refresh and try again

15.5 Report Generation Errors

Report times out:

- Reduce date range
- Reduce number of fields
- Export in smaller batches

No data in report:

- Check filter settings
- Verify data exists
- Clear report cache

15.6 Export Failures

Export fails:

- Reduce data volume
- Try different format
- Check file permissions
- Contact support

15.7 ATM Report Issues

Cannot create report:

- Verify ATM exists
- Check downtime reason selected
- Ensure open date valid

Report won't close:

- Check permissions
- Verify report is open
- Contact support

15.8 Computer Transfer Errors

Transfer fails:

- Verify target branch exists
- Check computer status
- Ensure user has transfer permissions
- Check for pending maintenance

15.9 Permission Errors

Access denied:

- Verify user role
- Check specific permissions
- Contact administrator

Cannot perform action:

- Request permission from administrator
- Verify role assignment

15.10 Activity Log Not Loading

Log won't display:

- Reduce date range
- Clear filters
- Check permissions
- Contact support

15.11 Mobile Access Issues

Mobile site not responsive:

- Update browser
- Clear cache
- Check orientation
- Contact support

15.12 Browser Compatibility Issues

Features not working:

- Update browser
- Try different browser

- Clear cache/cookies
- Disable extensions

15.13 Network Connectivity

Cannot connect:

- Check internet connection
- Verify VPN (if required)
- Contact network team
- Check maintenance schedule

15.14 Database Errors

Database connection errors:

- Contact system administrator
- Check maintenance schedule
- Wait and retry

15.15 Error Messages

Common error messages:

Message	Meaning	Action
"Record not found"	Data doesn't exist	Verify ID, refresh
"Duplicate entry"	Record already exists	Use unique value
"Permission denied"	Insufficient rights	Request access
"Validation failed"	Invalid input	Check field formats
"Session expired"	Logged out	Login again

15.16 Error Codes Reference

Code	Description	Solution
400	Bad request	Check input
401	Unauthorized	Login again
403	Forbidden	Check permissions
404	Not found	Verify URL/ID
422	Validation error	Check fields
500	Server error	Contact support
503	Service unavailable	Try later

15.17 Escalation Procedures

Level 1 – Self-service:

- Try solutions in this manual
- Check FAQs
- Clear cache, retry

Level 2 – Department support:

- Contact branch IT
- Create support ticket

Level 3 – Central IT:

- Escalate through help desk
- Provide error details
- Include screenshots

15.18 Support Ticket Creation

To create a support ticket:

1. Navigate to **Help** → **Support**
2. Click **New Ticket**
3. Enter:
 - Subject
 - Description
 - Priority
 - Attachments
4. Click **Submit**
5. Track ticket status

Section 15: Best Practices

16.1 Data Entry Standards

- Use consistent formats
- Complete all required fields
- Verify information before saving
- Use meaningful descriptions
- Follow naming conventions

16.2 Naming Conventions

ATMs: Use official terminal ID and location name

Computers:

- Tag: DB/[CODE]/4.1/[NUMBER]
- Host: W-BRN-[CODE]-[NUMBER]

Branches: Use official branch names

16.3 Tag Number Formats

Computer tag format:
DB/[BRANCH_TAG]/4.1/[SEQUENCE]

Examples:

- DB/JM/4.1/17591 – Jimma Branch
- DB/TBM/4.1/10964 – Tabot Maderia Branch

16.4 Regular Updates

- Update asset status promptly
- Close ATM reports when resolved
- Update computer specifications after upgrades
- Review branch information quarterly

16.5 Asset Tracking

- Scan asset tags regularly
- Verify asset locations
- Update assignments immediately

- Report discrepancies

16.6 Report Generation Schedule

Report	Frequency
ATM Status	Daily
Computer Inventory	Weekly
Asset Transfer	Monthly
Maintenance Summary	Monthly
Compliance Report	Quarterly

16.7 User Training

- New users: Complete training before access
- Annual refresher training
- Document process changes
- Share best practices

16.8 Security Best Practices

- Strong passwords
- Never share credentials
- Logout when away
- Report suspicious activity
- Keep contact information current

16.9 Password Policies

- Minimum 8 characters
- Mix of uppercase, lowercase, numbers, symbols
- Change every 90 days
- Don't reuse last 5 passwords
- Don't share password

16.10 Session Management

- Logout when finished
- Don't leave sessions open
- Use private/incognito mode on shared computers
- Session timeout after 15 minutes idle

16.11 Data Backup

- System backups automated
- Export critical reports locally
- Document recovery procedures

16.12 Audit Compliance

- Review activity logs monthly
- Document all changes
- Maintain audit trail
- Cooperate with auditors

16.13 Performance Optimization

- Use filters for large datasets
- Export rather than print
- Archive old data
- Clear browser cache regularly

16.14 Regular Maintenance

- Update browser versions
- Clear cache weekly
- Review system settings
- Test backup restoration

Section 16: Appendices

17.1 Appendix A: Field Formats

Field	Format	Example
ATM Terminal ID	PATML#### or PAGRL####	PATML059
ATM IP Address	IPv4	172.21.16.53
Computer Tag	DB/[CODE]/4.1/[NUMBER]	DB/JM/4.1/17591
Computer Host	W-BRN-[CODE]-[NUMBER]	W-BRN-JM-17591
Branch Code	Numeric	21
Branch Tag	2-4 letters	JM
Date	YYYY-MM-DD	2026-02-25
DateTime	YYYY-MM-DD HH:MM:SS	2026-02-25 14:30:00

17.2 Appendix B: Status Definitions

Status	Definition
Operational	Working normally
Non-Operational	Not working, needs attention
Under Maintenance	Sent for repair
Disposed	Removed from inventory
Relocating	Being moved
Online	ATM connected and working
Offline	ATM disconnected
Active	User account active
Inactive	User account disabled

17.3 Appendix C: ATM Type Codes

Code	Meaning
PAGRL	GRG ATM
PATML	NCR ATM (Lobby)
PATMT	NCR ATM (TTW)
PAGRR	GRG Recycler

17.4 Appendix D: Computer Models

Model ID	Model Name	Type
11	HP ProDesk 400 G4	Desktop
13	HP ProDesk 600 G2	Desktop
60	HP EliteDesk 800 G4	Desktop
61	HP EliteBook 840 G6	Laptop

62	HP ProBook 450 G5	Laptop
63	HP Compaq Pro 6300	Desktop
65	HP Compaq 8200 Elite	Desktop
66	HP Compaq 6200 Pro	Desktop
67	HP Compaq 6000 Pro	Desktop
72	HP Pro 3400 Series	Desktop
88	HP Compaq dc5800	Desktop

17.5 Appendix E: Branch Codes

Branch	Code	Tag
Jimma Branch	21	JM
Agaro Branch	22	AGR
Limmugenet Branch	23	LMG
Arbaminch Branch	052	AM
Wolaita Sodo Branch	043	WST
Hossaena Branch	051	HS
Jinka Branch	081	JK

17.6 Appendix F: District Reference

ID	Name	Code
1	Jimma District Office	JDO
2	South West District Office	SWDO
3	Nekemte District Office	NDO
4	Hawasa District Office	HDO
5	Adama District Office	ADO
6	Dire Dawa District Office	DDDO
7	Mekele District Office	MDO
8	Dessie District Office	DDO
9	Bahir Dar District Office	BDO
10	North Addis District Office	NADO
11	East Addis District Office	EADO
12	West Addis District Office	WADO
13	South Addis District Office	SADO
14	Wolaita District Office	WDO
15	Head Office	HO

17.7 Appendix G: Permission Matrix

Permission	Admin	Manager	Technician	Viewer
atm.create	✓	✓	✓	
atm.update	✓	✓	✓	
atm.delete	✓			
atm.view	✓	✓	✓	✓
computer.create	✓	✓	✓	
computer.update	✓	✓	✓	
computer.delete	✓			
computer.view	✓	✓	✓	✓
branch.create	✓			
branch.update	✓	✓		
branch.delete	✓			
branch.view	✓	✓	✓	✓
user.create	✓			
user.update	✓	✓		
user.view	✓	✓	✓	

17.8 Appendix H: Error Codes

Code	Description	Troubleshooting
400	Bad Request	Check input parameters
401	Unauthorized	Login required
403	Forbidden	Insufficient permissions
404	Not Found	Resource doesn't exist
409	Conflict	Duplicate entry
422	Unprocessable Entity	Validation failed
429	Too Many Requests	Slow down, wait
500	Internal Server Error	Contact support
503	Service Unavailable	System maintenance

17.9 Appendix I: Keyboard Shortcuts

Shortcut	Action
Alt + A	New ATM
Alt + C	New Computer
Alt + B	New Branch
Alt + R	New Report
Alt + D	Dashboard
Alt + S	Settings
Ctrl + K	Global Search
Ctrl + S	Save
Ctrl + F	Find/Search
Esc	Cancel/Close

17.10 Appendix J: Glossary of Terms

Term	Definition
ATM	Automated Teller Machine
Branch Code	Unique identifier for each branch
District Office	Regional administrative hub
Downtime	Period when an ATM is non-operational
GRG	GRG Banking – ATM manufacturer
Host Name	Network identifier for a computer
IFB	Interest-Free Banking
NCR	NCR Corporation – ATM manufacturer
Tag Number	Unique asset identifier
Terminal ID	Unique ATM identifier
TT	Trouble Ticket – support reference
TTW	Through-the-Wall ATM design

17.11 Appendix K: Frequently Asked Questions

Q: How do I reset my password?

A: Click "Forgot Password" on login screen or contact your administrator.

Q: How do I report an ATM issue?

A: Navigate to ATMs → Create Report, select the ATM, choose reason and action.

Q: How do I transfer a computer to another branch?

A: Find the computer, click Transfer, select destination branch, confirm.

Q: Why can't I see certain menu items?

A: You may not have the required permissions. Contact your administrator.

Q: How often is data backed up?

A: System backups run daily. Critical data is backed up in real-time.

Q: Can I access EMS on my phone?

A: Yes, EMS is mobile-responsive. Use any modern mobile browser.

17.12 Appendix L: Release Notes

Version 1.0 (February 2026)

- Initial release
- Core modules: ATMs, Computers, Branches, Assets
- Activity logging
- User management
- Reporting

17.13 Appendix M: System Requirements

Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5
RAM	4 GB	8 GB
Storage	100 MB	500 MB
Display	1366 x 768	1920 x 1080
Network	1 Mbps	5 Mbps
Browser	Chrome 90+	Latest Chrome

17.14 Appendix N: Training Materials

Available training resources:

- Video tutorials ([link](#))
- Quick reference cards ([link](#))
- Online help system
- Instructor-led training (by request)

17.15 Appendix O: Quick Reference Cards

ATM Quick Reference:

1. View ATMs → All ATMs
2. Create report → Select ATM → Choose reason → Submit
3. Close report → Find open report → Close

Computer Quick Reference:

1. Add computer → Fill details → Save
2. Transfer → Select computer → Choose branch → Confirm
3. Assign → Select computer → Choose user → Assign

Report Quick Reference:

1. Navigate to Reports
2. Select report type
3. Set filters
4. Generate
5. Export as needed

Document Information

Field	Value
Document Title	IT Operation Support System (ITOSS) User Manual
Version	1.0
Date	February 25, 2026
Document ID	EMS-UM-2026-001
Classification	Internal Use Only
Author	EMS Documentation Team
Owner	IT Infrastructure Department
Location	\fileservers\docs\EMS\Manual
File Name	EMS_User_Manual_v1.0.pdf